

A3: YOLO

Olivia Johnson

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1 Introduction

Object detection plays an important role in autonomous driving and intelligent transportation systems. Detecting small objects such as traffic signs or traffic lights from vehicle-mounted cameras can be challenging due to their small size, distance, and occlusion.

In this assignment, the YOLO (You Only Look Once) object detection framework was used to detect traffic signs from road images. A pre-trained YOLOv8 model was evaluated and then fine-tuned using a traffic sign dataset. Different configurations were explored to improve small object detection performance.

2 Dataset Description and Preprocessing

The dataset used in this project was the **Mapillary Traffic Sign Dataset (MTSD)**, which contains images of traffic scenes with annotated traffic signs.

A subset of the dataset was used for training and evaluation. The annotations were originally provided in JSON format and were converted to the YOLO format:

```
<class_id> <x_center> <y_center> <width> <height>
```

All bounding box coordinates were normalized relative to the image dimensions.

The dataset was split into training, validation, and test sets:

- Training images: 9933
- Validation images: 1241
- Test images: 1243

Traffic sign classes were grouped into 12 categories including stop, yield, speed limit, warning, pedestrian crossing, and others.

3 Baseline YOLO Model Performance

A pre-trained YOLOv8 nano model (`yolo8n.pt`) was first evaluated on the test dataset without additional training.

The baseline model was able to detect some traffic signs but struggled with smaller and distant signs. Some objects were missed due to their small size, and false detections occurred in cluttered scenes.

4 YOLO Model Training and Experiments

The YOLOv8 model was fine-tuned using the prepared dataset. Two different configurations were tested to improve small object detection. Both of these models were trained using only 12 labels. You will see in the results that they do not detect the vehicles, which are detected in the pre-trained model. This is because I did not include those labels in my model.

4.1 Experiment 1: Image Size 640

The first experiment used the following parameters:

- Model: YOLOv8n
- Image size: 640
- Batch size: 16
- Epochs: 30

This configuration trained relatively quickly but struggled to detect very small traffic signs.

4.2 Experiment 2: Image Size 1280

The second experiment increased the image resolution to improve detection of small objects.

- Model: YOLOv8n
- Image size: 1280
- Batch size: 8
- Epochs: 30

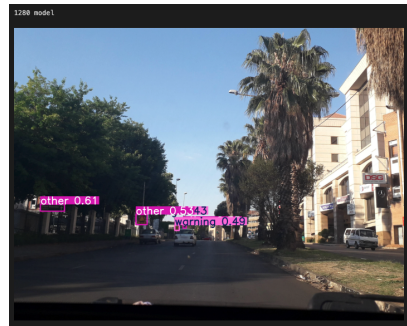
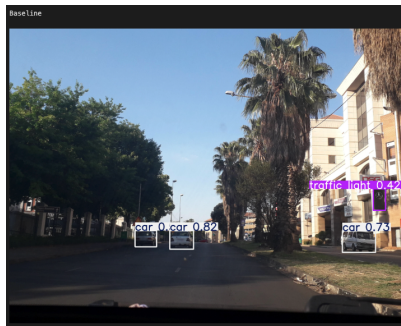
Higher resolution allowed the model to capture finer details and improved detection performance for distant signs.

5 Experimental Results

The performance of the models was evaluated using validation metrics such as precision, recall, and mean Average Precision (mAP).

	mAP50	mAP50-95	precision	recall
Pretrained YOLOv8n	0.010	0.006	0.017	0.071
YOLOv8n (640)	0.272	0.188	0.428	0.289
YOLOv8n (1280)	0.510	0.383	0.596	0.487

The model trained with larger image resolution showed improved performance in detecting small traffic signs.



6 Video or Image Sequence Testing

The trained model was also applied to video or image sequences to simulate real-world driving scenarios.

Frames with detection confidence above 0.5 were saved and analyzed. Some signs were missed however some signs were detected. I believe I would have had better results with a video clip that was less blurry.



7 Challenges in Small Object Detection

Detecting small objects presents several challenges:

- Small objects occupy very few pixels in the image.
- Objects may be far from the camera.
- Occlusions from other vehicles or objects.
- Class imbalance in the dataset.

These challenges can cause both missed detections and false positives.

8 Suggestions for Improving Detection

Several improvements could further enhance small object detection performance:

- Training for more epochs.
- Using a larger YOLO model (e.g., YOLOv8m).
- Increasing dataset size or balancing class distribution.
- Applying stronger data augmentation.
- Using higher image resolutions.

9 Conclusion

This assignment demonstrated the use of YOLO for detecting small traffic signs in road images. The pre-trained model provided a baseline performance, while fine-tuning on a traffic sign dataset improved detection accuracy.

Increasing image resolution significantly improved the detection of small objects. Overall, the trained YOLO model showed better performance compared to the baseline model.